Week 1 - Pre-class exercises

August 2026

These exercises let you practice the notions introduced in the pre-class notes. If an exercise resists after a few minutes, mark it and bring it to class.

> **Conventions used**
>
> - $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{R}_+ = [0, +\infty)$.
> - A data set with n observations and p variables is stored in a matrix
>
> $$X = (x_{ij}) \in \mathbb{R}^{n \times p},$$
>
> where x_{ij} is the value of variable j for observation i.

Exercise 1: Reading and writing symbolic statements

Translate each expression into precise English.

1. $x = 12$.
2. $c \neq 0$.
3. $n \leq 50$.
4. $r > 0$.
5. $0 \leq p_i \leq 1$ for every $i \in \{1, \ldots, n\}$.
6. $\mathbf{x} \in \mathbb{R}^p$.
7. $-2 \notin \mathbb{N}$.
8. $\forall i \in \{1, \ldots, n\}, \ e_i \geq 0$.
9. $\exists j \in \{1, \ldots, m\}$ such that $y_j < 0$.
10. $\sum_{j=1}^{p} w_j = 1$.

> **Solution**
>
> 1. x is equal to 12.
> 2. c is different from 0.
> 3. n is less than or equal to 50.
> 4. r is strictly greater than 0.
> 5. For every index i from 1 to n, the number p_i lies between 0 and 1, endpoints included.
> 6. The vector $\mathbf{x}$ has p real components.

7. -2 does not belong to the set of natural numbers.
8. For every index i, the number e_i is non-negative.
9. There exists at least one index j such that y_j negative.
10. The sum of weights w_j where j ranges from 1 to p, is equal to 1.

Exercise 2: Number sets, vectors, and matrices

State whether each proposition is true or false, and justify briefly. In proposition 8, write

$$M = \begin{pmatrix} 1 & 2 & 0 \\ 0 & \frac{1}{2} & 4 \end{pmatrix}.$$

1. $0 \in \mathbb{N}$.
2. $-3 \in \mathbb{Z}$.
3. $\frac{1}{4} \in \mathbb{Q}$.
4. $\sqrt{2} \in \mathbb{Q}$.
5. $\pi \in \mathbb{R}$.
6. $-1 \in \mathbb{R}_+$.
7. $(2, -1, 4) \in \mathbb{R}^3$.
8. $M \in \mathbb{R}^{2\times 3}$.
9. $(0.2, 0.5, 0.3) \in \mathbb{R}_+^3$.
10. If $X \in \mathbb{R}^{100\times 8}$ and the rows are observations, then the data set contains 100 variables.

Solution

Proposition	Answer	Justification
$0 \in \mathbb{N}$	True	$\mathbb{N}$ contains 0.
$-3 \in \mathbb{Z}$	True	The integer set include the negative integers too.
$\frac{1}{4} \in \mathbb{Q}$	True	It is a ratio of two integers with non-zero denominator.
$\sqrt{2} \in \mathbb{Q}$	False	$\sqrt{2}$ is irrational.
$\pi \in \mathbb{R}$	True	π is real because all irrational numbers are real numbers.
$-1 \in \mathbb{R}_+$	False	$\mathbb{R}_+$ contains only non-negative reals.
$(2, -1, 4) \in \mathbb{R}^3$	True	All three components are real.
$M \in \mathbb{R}^{2\times 3}$	True	The matrix has two rows and three columns.
$(0.2, 0.5, 0.3) \in \mathbb{R}_+^3$	True	All three components are non-negative real numbers.
100 variables	False	There are 100 observations and 8 variables.

Remark. The vector in proposition 9 has non-negative entries summing to 1. This is a vector of weights. The set of all such vector is called a **simplex** of size p, where p is the number of elements in the vector, and is defined as follows

$$\Delta_p = \left\{ \mathbf{w} \in \mathbb{R}^p_+ \;\middle|\; \sum_{j=1}^{p} w_j = 1 \right\}$$

So the vector in proposition 9 is a simplex of size 3. A probability vector is an example of a simplex vector.

Exercise 3: Set operations and data filters

A customer database is indexed by the universal set

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\} .$$

Define

- $A = \{1, 2, 4, 7\}$ as the customers whose annual expenditure d exceeds \$1,000; and
- $B = \{2, 3, 4, 6\}$ as the customers whose risk score s is at least 0.7.

Then:

1. Determine $A \cap B$, $A \cup B$, $A - B$, and $B - A$.
2. Determine the complement A^c relative to U.
3. Interpret $A \cap B$ in the business context.
4. Write, in set-builder notation, the set of observations satisfying both $d_i > 1000$ and $s_i \geq 0.7$.

Solution

1. $$A \cap B = \{2, 4\} , \qquad A \cup B = \{1, 2, 3, 4, 6, 7\} ,$$
$$A - B = \{1, 7\} , \qquad B - A = \{3, 6\} .$$
2. $$A^c = U - A = \{3, 5, 6, 8\} .$$
3. $A \cap B$ is the set of customers who simultaneously spend more than \$1,000 and carry a risk score of at least 0.7. That is, accounts who are both high-value and high-risk at the same time. Mathematically:
4. $$\{i \in U \mid d_i > 1000 \;\text{ and }\; s_i \geq 0.7\} = A \cap B.$$

Exercise 4: Absolute value

Compute:

1. $|-7|$;

2. $|3-8|$;
3. $|-\frac{1}{2}|$;
4. $\Big||-4|-6\Big|$.

Solution

$$|-7| = 7, \qquad |3-8| = |-5| = 5, \qquad |-\tfrac{1}{2}| = \tfrac{1}{2}.$$

For the last one, we work inside out:

$$\Big||-4|-6\Big| = |4-6| = |-2| = 2.$$

Exercise 5: Matrix representation of a data set

Consider

$$X = \begin{pmatrix} 2 & 5 & 1 \\ 4 & 3 & 0 \\ 6 & 7 & 1 \\ 8 & 2 & 0 \end{pmatrix}.$$

1. To which space does X belong? How many observations and how many variables does it record?
2. Write down the second observation.
3. Write down the third variable.
4. What is x_{32}, and what does it represent?
5. Using summation notation, write the sum of variable j over all observations.
6. Using summation notation, write the mean of variable j.

Solution

1. $X \in \mathbb{R}^{4\times 3}$. It contains 4 observations and 3 variables.
2. The second observation is the second row: $\mathbf{x}_2 = (4, 3, 0)$.
3. The third variable is the third column:

$$\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

4. $x_{32} = 7$. It represents the value of variable 2 for observation 3.
5. $$\sum_{i=1}^{n} x_{ij}.$$
6. $$\bar{x}_{\cdot j} = \frac{1}{n}\sum_{i=1}^{n} x_{ij}.$$

Exercise 6: Expanding and condensing summations

A. Write each sum without the symbol Σ.

1. $\sum_{i=1}^{5} x_i$.
2. $\sum_{t=0}^{3} r_t$.
3. $\sum_{j=2}^{5}(2j-1)$.
4. $\sum_{i=1}^{4} p_i q_i$.
5. $\sum_{k=1}^{n} c$, where c is a constant.

B. Rewrite each expression using the symbol Σ.

1. $a_1 + a_2 + \cdots + a_8$.
2. $x_1^2 + x_2^2 + \cdots + x_n^2$.
3. $2y_1 + 2y_2 + \cdots + 2y_m$.
4. $p_1q_1 + p_2q_2 + \cdots + p_nq_n$.
5. $(z_1-\mu)^2 + \cdots + (z_n-\mu)^2$.

> Solution
>
> **A. Expansion**
>
> 1. $x_1 + x_2 + x_3 + x_4 + x_5$.
> 2. $r_0 + r_1 + r_2 + r_3$.
> 3. $(2 \cdot 2 - 1) + (2 \cdot 3 - 1) + (2 \cdot 4 - 1) + (2 \cdot 5 - 1)$.
> 4. $p_1q_1 + p_2q_2 + p_3q_3 + p_4q_4$.
> 5. $\underbrace{c + c + \cdots + c}_{n \text{ terms}} = nc$.
>
> **B. Condensation**
>
> $$\sum_{i=1}^{8} a_i, \qquad \sum_{i=1}^{n} x_i^2, \qquad \sum_{i=1}^{m} 2y_i = 2\sum_{i=1}^{m} y_i,$$
> $$\sum_{i=1}^{n} p_i q_i, \qquad \sum_{i=1}^{n}(z_i-\mu)^2.$$

Exercise 7: Price index

A basket contains three goods, with quantities, base-year prices, and year-t prices

$$\mathbf{q} = (10, 5, 2), \qquad \mathbf{p}^0 = (2, 4, 10), \qquad \mathbf{p}^t = (2.5, 5, 12).$$

1. Compute the cost of the basket in the base year, $C_0 = \sum_{i=1}^{3} p_i^0 q_i$.
2. Compute the cost of the same basket in year t, $C_t = \sum_{i=1}^{3} p_i^t q_i$.
3. Compute the index $I_t = \dfrac{C_t}{C_0} \times 100$.
4. Interpret the result.

Solution

1. $$C_0 = 2(10) + 4(5) + 10(2) = 20 + 20 + 20 = 60.$$
2. $$C_t = 2.5(10) + 5(5) + 12(2) = 25 + 25 + 24 = 74.$$
3. $$I_t = \frac{74}{60} \times 100 \approx 123.33.$$
4. The basket costs about 123.33% of what it cost in the base year. That is, its cost has increased by roughly 23.33%.

Exercise 8: Prediction errors and loss functions

Observed values and predictions are

$$\mathbf{y} = (10, 13, 15, 20), \qquad \hat{\mathbf{y}} = (11, 12, 14, 22).$$

1. Compute the residuals $e_i = \hat{y}_i - y_i$.
2. Compute the mean absolute error
$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |e_i|$$
.
3. Compute the mean squared error
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} e_i^2$$
.
4. Compute the root mean squared error $\text{RMSE} = \sqrt{\text{MSE}}$.
5. Which observation contributes most to the MSE? Explain.
6. Recompute the MAE and the MSE when the last prediction changes from 22 to 25. By what factor does each one increase? Which of the two reacts more strongly to the change, and why?

Solution

1. $$\mathbf{e} = \hat{\mathbf{y}} - \mathbf{y} = (11 - 10,\ 12 - 13,\ 14 - 15,\ 22 - 20) = (1, -1, -1, 2).$$
2. $$\text{MAE} = \frac{1}{4}(|1| + |-1| + |-1| + |2|) = \frac{5}{4} = 1.25.$$
3. $$\text{MSE} = \frac{1}{4}(1^2 + (-1)^2 + (-1)^2 + 2^2) = \frac{7}{4} = 1.75.$$
4. $$\text{RMSE} = \sqrt{1.75} \approx 1.323.$$
5. Observation 4 because its squared error is $2^2 = 4$, whereas each of the other squared errors is 1.

6. With $\hat{y}_4 = 25$ the residuals become $(1, -1, -1, 5)$, so

$$\text{MAE}_{\text{new}} = \frac{1+1+1+5}{4} = 2, \qquad \text{MSE}_{\text{new}} = \frac{1+1+1+25}{4} = 7.$$

The MAE went from 1.25 to 2 (a factor of 1.6), the MSE from 1.75 to 7 (a factor of 4). The MSE increases much more rapidly because squaring magnifies the influence of large errors.

Exercise 9: Optimization modelling (allocating a marketing budget)

A company splits a budget across three digital channels. Let x_j be the amount, in thousands of dollars, invested in channel j. The estimated incremental profit, in thousands of dollars, is

$$7x_1 + 5x_2 + 4x_3.$$

The following is known:

- the total budget cannot exceed 100;
- campaign-management capacity imposes $2x_1 + x_2 \leq 140$;
- risk exposure must satisfy $0.4x_1 + 0.2x_2 + 0.1x_3 \leq 25$;
- at least 10 must be invested in the third channel;
- investments cannot be negative.

Then:

1. Identify the decision variables, the parameters, the objective function, and the constraints.
2. Write the feasible set D for the decision variables in set-builder notation.
3. Write the optimization problem in full.
4. Determine whether the decisions $(40, 30, 20)$ and $(60, 20, 20)$ are feasible.

Solution

1. the decision variables are x_1, x_2, x_3 (the amounts invested). Recall that a parameter is a number given to us before solving, that is, all the numbers in the exercise statement is a parameter. That is, the unit profits $7, 5, 4$; The capacities 100 and 140; the coefficients of the constraints for the campaign management capacity $(2, 1)$ risk exposure $(0.4, 0.2, 0.1)$; and the limits 100, 140, 25, and 10. The objective function is $7x_1 + 5x_2 + 4x_3$, which aim to maximize. Finally, the constraints are the 5 bullet points in the exercise statement.

2. $$D = \left\{ \mathbf{x} \in \mathbb{R}^3_+ \;\middle|\; \begin{aligned} x_1 + x_2 + x_3 &\leq 100, \\ 2x_1 + x_2 &\leq 140, \\ 0.4x_1 + 0.2x_2 + 0.1x_3 &\leq 25, \\ x_3 &\geq 10 \end{aligned} \right\}.$$

   Writing $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3_+$. Notice that the feasible set is the set of numbers satisfying all 5 constraints.

3. $$\max_{x_1,x_2,x_3} \quad 7x_1 + 5x_2 + 4x_3$$

   subject to

   $$\begin{cases} x_1 + x_2 + x_3 \leq 100, \\ 2x_1 + x_2 \leq 140, \\ 0.4x_1 + 0.2x_2 + 0.1x_3 \leq 25, \\ x_3 \geq 10, \\ x_1, x_2, x_3 \geq 0. \end{cases}$$

4. For $(40, 30, 20)$, the budget used is $40 + 30 + 20 = 90 \leq 100$; the capacity expression is $2(40) + 30 = 110 \leq 140$; the risk expression is $0.4(40) + 0.2(30) + 0.1(20) = 24 \leq 25$; $x_3 = 20 \geq 10$; and $(40, 30, 20) \in \mathbb{R}^3_+$. Hence, all five constraints hold and the decision is feasible.

   For $(60, 20, 20)$, the risk expression is

   $$0.4(60) + 0.2(20) + 0.1(20) = 30 > 25$$

   which violates one of the constraints. Violating one of the constraints is enough to reject the decision.

Exercise 10: System of linear inequalities

In an optimization problem the objective is to minimize

$$3t + 2u,$$

subject to

$$t \geq |2x - 3y + 5|, \qquad u = \max\{0,\ 4x - y - 6\}.$$

1. Replace these two relations by a system of linear inequalities.
2. Compute t, u, and $3t + 2u$ for $x = 2$ and $y = 1$.

Solution

1. A number is $\geq |a|$ when it is both $\geq a$ and $\geq -a$, so

   $$t \geq |2x - 3y + 5| \iff \begin{cases} t \geq 2x - 3y + 5, \\ t \geq -(2x - 3y + 5). \end{cases}$$

   The relation $u = \max\{0,\ 4x - y - 6\}$ is replaced by

   $$\begin{cases} u \geq 0, \\ u \geq 4x - y - 6. \end{cases}$$

2. Because we are minimizing the positive sum of t and u, we compute the smallest t and u such that the inequalities above hold. The smallest admissible value are achieved at the

equality "=". Hence, at $x = 2$ and $y = 1$ we have,

$$t = |2(2) - 3(1) + 5| = |6| = 6,$$

and

$$u = \max\{0,\ 4(2) - 1 - 6\} = \max\{0, 1\} = 1,$$

so

$$3t + 2u = 3(6) + 2(1) = 20.$$

Exercise 11: Isolating a variable using the log function

A multiplicative relation is given by

$$Y = AX^{\beta}e^{-\gamma Z},$$

where $A > 0$, $X > 0$, $Y > 0$, and $\beta \neq 0$.

1. Isolate X as a function of Y, A, β, γ, and Z.
2. Transform the relation into an equation that is linear in $\ln X$ and Z.
3. Compute X when $A = 12$, $\beta = 0.8$, $\gamma = 0.15$, $Z = 4$, and $Y = 30$.

Solution

1. Divide by A and multiply by $e^{\gamma Z}$:

$$X^{\beta} = \frac{Ye^{\gamma Z}}{A}.$$

Because the right-hand side is strictly positive and $\beta \neq 0$, we can raise both sides to the power $1/\beta$:

$$X = \left(\frac{Ye^{\gamma Z}}{A}\right)^{1/\beta}.$$

2. Now that we have the power, taking the natural log of each side allows us to turn the multiplications into a linear sum:

$$\ln Y = \ln A + \beta \ln X - \gamma Z.$$

3. Substituting,

$$X = \left(\frac{30\,e^{0.15\times 4}}{12}\right)^{1/0.8} = \left(\frac{30\,e^{0.6}}{12}\right)^{1.25} \approx (4.5553)^{1.25} \approx 6.655.$$

Exercise 12: log-linear form

Consider the model

$$Y = AX^{\beta}e^{\gamma Z},$$

where $A > 0$ and $X, Y > 0$. Three transformed observations are available:

Observation	$\ln X$	Z	$\ln Y$
1	0	0	1
2	1	0	3
3	0	2	0

1. Write the log-linear form of the model.
2. Identify A, β, and γ exactly.
3. Predict Y when $X = e^2$ and $Z = 1$.

Solution

1. Taking natural logarithms:

$$\ln Y = \ln A + \beta \ln X + \gamma Z.$$

   Set $\alpha = \ln A$, so that $\ln Y = \alpha + \beta \ln X + \gamma Z$.
2. Substituting the three observations give us

$$\begin{cases} \alpha + \beta(0) + \gamma(0) = \alpha = 1, \\ \alpha + \beta(1) + \gamma(0) = \alpha + \beta = 3, \\ \alpha + \beta(0) + \gamma(2) = \alpha + 2\gamma = 0. \end{cases}$$

   The first equation gives $\alpha = 1$; substituting into the second gives $\beta = 2$; substituting into the third gives $\gamma = -\frac{1}{2}$. Since $A = e^{\alpha}$,

$$A = e, \qquad \beta = 2, \qquad \gamma = -\tfrac{1}{2}.$$

3. With $X = e^2$ we have $\ln X = 2$, and $Z = 1$, so

$$\ln Y = 1 + 2(2) - \tfrac{1}{2}(1) = 1 + 4 - \tfrac{1}{2} = \tfrac{9}{2},$$

   hence

$$Y = e^{9/2} \approx 90.02.$$